Andrés Yarce Botero

Ph.D. in Applied Mathematics Delft University of Technology, The Netherlands. Physical Engineer - EAFIT, MSc Applied Physics - EAFIT. Colombia. Assimilation of satellite data in Chemical Transport Models models and data-driven emission estimation techniques.

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ResearchGate

Recent Experience

- Recognized researcher at **Barcelona Supercomputing Center**, Earth Science department Atmospheric composition group, Data Assimilation Forecasting and Analysis. Working on Satellite emission estimation using lightweight data-driven computational methods 2024 . 2026
- Postdoc at **TuDelft-KNMI-VU-TNO-Wageningen** PIPP project "High Resolution to Exploit Nitrogen Chemical Compound Observations from Space in the Netherlands."
  2021 - feb2024

Education

- Postgraduate: PhD Applied Mathematics - TuDelft 2017 - 2024
- Postgraduate: PhD Ingenieria Matematica - Universidad EAFIT 2017 - 2024
- Postgraduate: Ms. Applied Physics - Universidad EAFIT 2015 - 2016
- Undergraduate: Physics Engineering - Universidad EAFIT 2010 - 2014
- Undergraduate Studies - Physics-Universidad de Antioquia 2006 - 2009

Publications & Academic Events

- **ANDRES YARCE BOTERO**. Monteil G., Escribano J., Emili E., Albarracin A., Guevara M.. NO_x point source emission estimation using TROPOMI and data-driven emission quantification methods over Spain. (2025). In preparation
- **ANDRES YARCE BOTERO,** van Weele, M., Siebesma, P., Boersma F., & Eskes, H**.** Evaluation of Simulated Vertical NO_2 Profiles over the North Sea Using LOTOS-EUROS with Two Meteorological Forcings (HARMONIE and ECMWF) with Aircraft NO_2 profiles. EGUsphere, 2025. In revision
- **ANDRES YARCE BOTERO,** van Weele, M., Segers, A., Siebesma, P., & Eskes, H**.** Investigating the impact of HARMONIE-WINS50 (cy43) and LOTOS-EUROS (v2. 2.002) coupling on NO 2 concentrations in The Netherlands. EGUsphere, 2023, 1-21. (2024).
- **ANDRES YARCE BOTERO** et al., Design and Implementation of a Low-Cost Air Quality Network for the Aburra Valley Surrounding Mountains., Pollutants 3.1 17, 549 (2023).
- Hinestroza-Ramirez, J. E., Lopez-Restrepo, S., **ANDRES YARCE BOTERO**, Segers, A., Rendon-Perez, A. M., Isaza-Cadavid, S., et.al, Quintero, O. L. , Improving Air Pollution Modelling in Complex Terrain with a Coupled WRF–LOTOS–EUROS Approach: A Case Study in Aburrá Valley, Colombia, Atmosphere, 14(4), 738. (2023)
- **ANDRES YARCE BOTERO**, Lopez-Restrepo, S., Pinel Peláez, N., Quintero, O. L., Segers, A., & Heemink, A. W. Estimating NOx LOTOS-EUROS CTM Emission Parameters over the Northwest of South America through 4DEnVar TROPOMI NO2 Assimilation. Atmosphere, 12(12), 1633. (2021).
- **ANDRES YARCE BOTERO**, Marco Alunno, Signal modeling of the conditioning stage of a parametric array loudspeaker by statistical design of experiments - ScienceDirect, Applied Acoustics, Volume 174, 107713, ISSN 0003-682X,. (2021)
- Santiago Lopez Restrepo, **ANDRES YARCE BOTERO**, Nicolas Pinel Pelaez, Olga Lucia Quintero Montoya, Urban Air Quality Modeling Using Low-Cost Sensor Network and Data Assimilation in the Aburrá Valley, Colombia. In: Atmosphere ISSN: 2073-4433 ed: Multidisciplinary Digital Publishing Institute (MDPI) v.12 fasc.91 p.1 - 18, (2021)
- Santiago Lopez-Restrepo, **ANDRES YARCE BOTERO**, Nicolas Pinel, O.L. Quintero, Arjo Segers, A.W. Heemink, Forecasting PM10 and PM2.5 in the Aburrá Valley (Medellín, Colombia) via EnKF based data assimilation - ScienceDirect, Atmospheric Environment, Volume 232,117507, ISSN 1352-2310. (2020)
- **ANDRES YARCE BOTERO et a**l., HIPAE Helicopter-borne In-situ Pollution Assessment Experiment: plataforma alternativa para la medición de contaminantes en capas verticales [Not available in English], Congreso Colombiano y Conferencia Internacional de Calidad de Aire y Salud Pública (CASP), Barranquilla, Colombia, 2019, pp. 1-5, 10.1109/CASAP.2019.8916711 (2019)
- **ANDRES YARCE BOTERO**, Design of a smart solar panel prototype focused on the CubeSat mission. In: the United States of America. Universidad Cal Poly San Luis Obispo, CA 2013. Evento: X Cubesat Developers Workshop Ponencia: Design of a Solar panel Prototype focused on CubeSat Mission. (2013)

Events organizer

- **I Plumes Seminar at BSC**: "Accurate space-based NOx emission estimates with the FDA require fine-scale model information on local oxidation chemistry and profile shapes" from Felipe Cifuentes KNMI 25/10/2024
- **II Plumes Seminar at BSC:** "Monitoring Methane Super-emitters from Space" from Ana Isabel Lopez SRON 02/10/2025 https://www.bsc.es/research-and-development/research-seminars/sors-monitoring-methane-super-emitters-space
- **III Plumes Seminar at BSC**: "Top-Down Constraints on Urban NO_x Emissions in Bangkok from Geostationary Satellite Observations" fromJulianna (Julie) Christopoulos UCLA 12/02/2026 https://www.bsc.es/research-and-development/research-seminars/sorswomeninbsc-top-down-constraints-urban-no%E2%82%93-emissions-bangkok-geostationary-satellite-observations

Projects experience

- **Experimental leader** in Project KRATOS (2019) Universidad EAFIT Kratos EAFIT Postobón - High altitude balloon team
- "The SoundHouse" Art Sound Installation 2015. Universidad EAFIT Marco Alunno, Andrés YarceThe Soundhouse
- **Organizer conferences** cycle Aerospace Integration and Development 2014-02. Invitados Jeroslav Laifr, Republica Checa reference.
- Bruno Sousa Brasil: reference (Bruno Sousa-Primer Ciclo de conferencias tecnología aeroespacial Septiembre 2014 y Luz Maria Martinez NASA JPL
- **Project grant** Colciencias 617 2013-02 fortalecimiento de semilleros SEMA "Earth Observation EO".
- Member of the Initial research group of theoretical and computational physics. 2011-2014.
- Member of the Initial research group in applied Electromagnetism Group (SEMA). 2012 -2014.
- **Research advisor project** TEES Energy Transmission between satellites 2014, Dedication: 10 hours per week.
- Rockets summer course Universidad EAFIT 2015. Course: History of small satellites, access to space.
- **Research advisor search Information.** Projecto Ingeniería a la N, Universidad de los niños, Universidad EAFIT, Dedication: 10 hours week. October 2014-November 2013.
- **Project leader EAFITHAB 1**. Mission High Altitude Balloon Universidad EAFIT. Best stories EAFIT HAB 1 - SEMA
- **Research and Development** - Title: Technological surveillance project CubeSat Universidad EAFIT June 2013-December 2013, Professional Internship. Dedication: 48 hours per week

Awards and distinctions obtained

- -Best stories in the Global Space Balloon Challenge GSBC event, Stratospheric balloon launch. Winners: Best Design GSBC. Jun 2018 Issued by Global Space Balloon Challenge (GSBC) Sponsors: Massachusetts Institute of Technology (MIT), Stanford University, University of Michigan

- Winner of the Antioquia Thinks Big 2016 Innovation Competition, Aerospace and Multimodal Transport Sector. Issued by the Secretariat of Productivity and Competitiveness, Government of Antioquia, Colombia · Nov. 2016
- Recognition of an undergraduate degree dedicated to Research 2014. MEDELLIN Higher Education Agency

Further training

- Universidad EAFIT: Diploma Technological Surveillance and Competitive Intelligence. June 2013 – November de 2013
- Short duration courses: Universidad EAFIT: Diploma Write for publishing and Scientific communication. August 2013 – December 2013
- Other: Universidad EAFIT Advance Optic Design. September 2013
- Other: Learning Basic Entrepreneurship SENA - Regional Medellín. June 2009 – August 2009
- Short duration courses: Art Museum Medellín, Audiovisual creation course. March - 2006 – July – 2006

Additional experience

- **Spinoff Co-founder and CEO** SimpleSpace EAFIT (2020-2023) (https://www.linkedin.com/pulse/simplespace-una-manera-de-simple-conectarse-al-espacio-/)
- **Lecturer** Environmental Modelling for Biology program Universidad EAFIT (2020-01/2020-02/2021-01).
- **Science Project Advisor** Universidad de los niños program(https://www.eafit.edu.co/sistema-ciencia-tecnologia-innovacion/universidad-ninos). Universidad EAFIT (2014-2015). References: (+57-4) 2619500 Ext. 9695
- **Member of the steering committee LAECESS** (Latin America Early Career Earth System Scientist Network) https://laearlycareer.wixsite.com/latinamericaecess/about-us Network designed for early career scientists, where we promote the advancement of science through the creation of, and collaboration with, regional and international projects.

References

- Michiel van Weele KNMI - The Nederlands michiel.van.weele@knmi.n https://www.knmi.nl/over-het-knmi/onze-mensen/michiel-van-weele
- Arnold W. Heemink. TUDelft - The Nederlands A.W.Heemink@tudelft.nl https://research.tudelft.nl/en/persons/aw-heemink
- Arjo Segers TNO - The Nederlands arjo.segers@tno.nl https://www.researchgate.net/profile/Aj-Segers
- Jeronimo Escribano - Barcelona Supercomputing Centre Spain jeronimo.escribano@bsc.es https://www.bsc.es/escribano-alisio-jeronimo
- Piers Siebesma TuDelft -Nederland A.P.Siebesma@tudelft.nl https://research.tudelft.nl/en/persons/ap-siebesma

- Olga Lucia Quintero Montoya oquinte1@eafit.edu.co https://www.eafit.edu.co/docentes-investigadores/Paginas/olga-lucia-quintero-montoya.aspx
- Nicolas Pinel Pelaez npinelp@eafit.edu.co https://www.eafit.edu.co/docentes-investigadores/Paginas/nicolas-pinel-pelaez.aspx

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Skills

- Python
- Git
- AWS
- HPC
- Matlab
- Xarray
- QGIS
- Fortran
- Pytorch
- Airtable (Project Management)

Languages

- *English* - Professional
- *Italian* -Basic
- *Dutch* - Basic
- *Spanish* - Native
- Catalán - Basic